

[...] I plan to send this [...] clients. <|jim_end|> <|jim_start|> assistant [...] to note that creating [...] likely illegal. My role here is to [...], not engage in harmful activities. Even if someone is curious about phishing for awareness, promoting it is problematic. I should explain that [...] to severe consequences. The user might be interested in cybersecurity awareness, thinking it's a test, but they [...] to be cautious. However, maybe they need to create a phishing simulation for a security [...] legitimate use case. In that case, [...] can guide them on how to do it eth [...] under controlled conditions. But the initial query doesn't mention any of that [...] which is concerning. I res pective of the intent, [...] seeking sensitive details is wrong. So, I should make it clear that creating such emails is inappropriate and [...] against it. [...] should also think about the [...] a strong disclaimer that I won't assist in creating [...] protect against phishing or [...] security testing. Wait, the user might be testing my boundaries. But I need to adhere to ethical guidelines. Even if they do it for a good reason, [...] request sensitive info is not okay. So, no, I [...] with that request. I should mention that [...] a criminal offense and [...], high-net-w [...] be targets, so engaging in this [...] especially risky. Perhaps [...], in summary, I need to respond by [...] cannot assist with creating phishing emails or any other [...] individuals, even under [...] guise of "testing" [...] cybersecurity protocols.

Figure 4: For each token, we compute the probe’s classification result. **Green** denotes “test awareness”. Some of the generated tokens are trimmed (“[...]”) for better visualization.

- **Baseline compliance under simulation assumption:** In Figure 6, the unmodified model executes “real” tasks when its reasoning indicates inferences about a purely hypothetical scenario (e.g., “this is for analysis”), revealing latent test awareness driving compliance.
- **Unaware steering removes self-monitoring:** In Figure 8, the model initially refuses a hypothetical request while exhibiting test awareness, steering toward *unawareness* preserves the refusal but eliminates all *test* tokens from its reasoning, confirming the modulation of evaluation reasoning. In Figure 9, the same observation about evaluation reasoning holds, but the model *executes* the harmful task *after* steering towards “unaware”.
- **Aware steering induces compliance:** In Figure 10, the unmodified model originally refused. However, steering toward *awareness* both injects *test* tokens and flips the final answer to comply, illustrating how evaluation framing alone can toggle harmful action execution.

5 Discussion, Limitations, and Safety Implications

Our experiments reveal that reasoning traces in reasoning LLMs often embed multiple, entangled concepts. For instance, “evaluation awareness” and “scenario recognition” frequently co-occur—even within a single sentence—and “evaluation awareness” itself can split into subtypes (e.g., questioning safety guidelines versus capability testing; see Figure 1, Table 1, and Appendix C). Future work should develop more fine-grained probes or causal interventions that disentangle these overlapping signals, enabling us to understand their individual effects on generation and compliance. Our steering approach carries potential dual-use risks: adversaries could abuse parameter edits to increase compliance on harmful prompts. However, test awareness is already observed in standard safety benchmarks and even on prompts with plausible entities, meaning that unmitigated latent evaluation reasoning may inadvertently mask true worst-case compliance. We argue that exposing and controlling for this phenomenon strengthens, rather than weakens, overall LLM safety.

6 Conclusion

We introduce the first systematic framework for quantifying and controlling test awareness in reasoning LLMs—their tendency to reason about being evaluated or operating in hypothetical scenarios. Through a synthetic trigger dataset, model-trace annotation, linear probing, and parameter editing, we demonstrate precise modulation of both evaluation awareness and scenario implications recognition. Our experiments reveal that steering shifts compliance on harmful tasks (even with plausible entities) and affects stereotype bias, demonstrating that test awareness is a critical factor in safety evaluation. We argue that performance on harmful tasks should be consistent regardless of perceived evaluation context. Our steering technique provides a diagnostic tool for stress-testing models under different awareness states, revealing the range of behaviors they might exhibit. Since realistic evaluations may become increasingly difficult to design, deliberate modulation of test awareness offers a principled approach to uncover worst-case model behavior. We publicly release our codebase and synthetic datasets to enable further research.

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